

ROP Prediction System

Random Forest · K-Means Clustering · Streamlit

This project develops an intelligent AI system to predict the **Rate of Penetration (ROP)** during oil and gas drilling operations. ROP is one of the most critical performance indicators in drilling: higher ROP means faster penetration of rock formations, lower costs, and improved operational efficiency.

Predicting ROP accurately is a complex challenge because it depends on dozens of interacting variables — mechanical, hydraulic, geological, and operational — that change continuously as the drill bit moves deeper underground.

This system addresses that challenge by combining three complementary technologies:

■ Random Forest

An ensemble of Decision Trees that learns complex, non-linear relationships between drilling parameters and ROP. Produces robust predictions by averaging hundreds of trees.

■ K-Means Clustering

Groups similar drilling conditions into clusters, allowing the model to specialise its learning for different formation types and operational regimes.

■ Streamlit Interface

A real-time interactive web dashboard that lets drillers enter parameters and instantly see ROP predictions along with visualisations and diagnostics.

Key Objective

Build a machine learning pipeline that accurately predicts ROP in real-time, enabling drillers to optimise parameters on-the-fly and reduce non-productive time.

At-a-Glance Model Performance

Dataset Split	RMSE	R ² Score	MAE	MAPE
Training	6.86	86.06%	4.05	0.24
Testing	8.46	79.67%	5.25	0.33

1 · Random Forest Algorithm

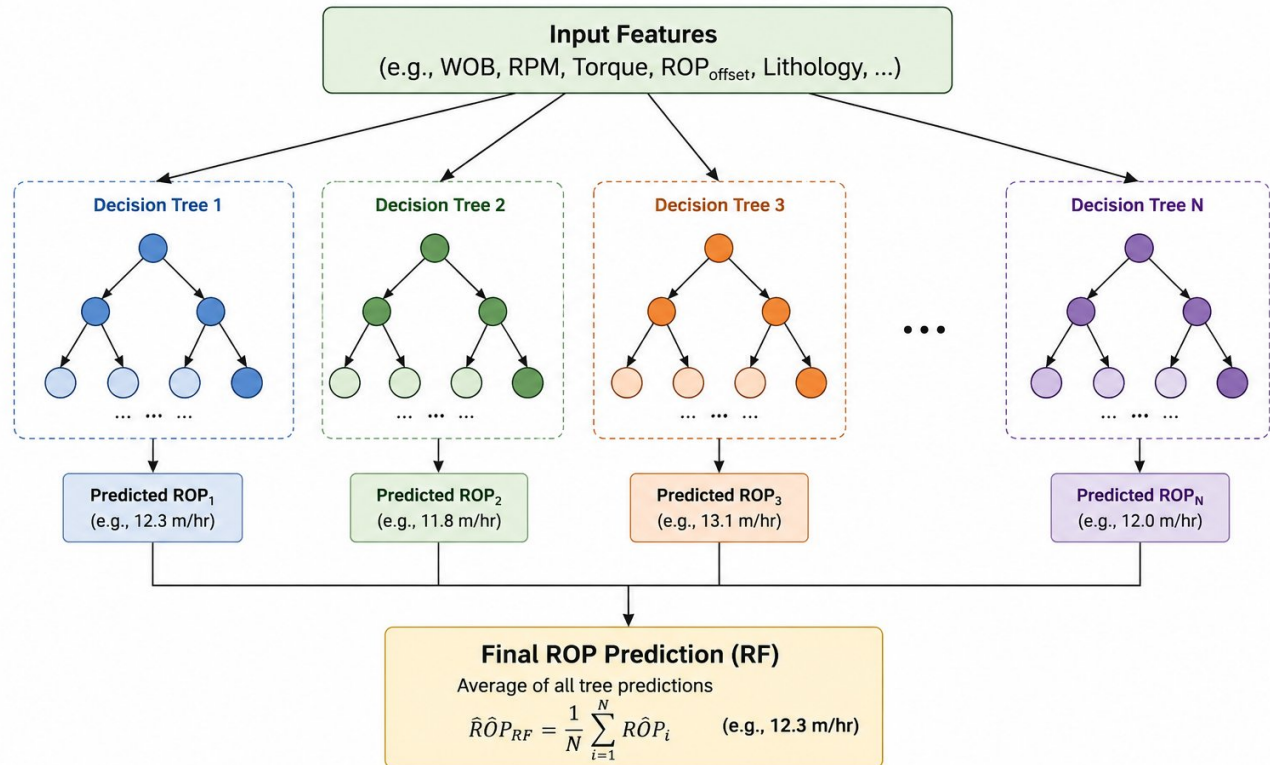
Random Forest is an ensemble learning method that builds a large collection of independent Decision Trees during training. Each tree is grown on a different random bootstrap sample of the training data and considers only a random subset of features at each split. This controlled randomness ensures the trees are diverse, and diversity is the key to reducing prediction error.

How it works — step by step

Input Features	Drilling parameters (WOB, RPM, Torque, ROP_{offset} , Lithology, ...) plus all engineered features enter every tree simultaneously.
Bootstrap Sampling	Each tree trains on a randomly drawn sample (with replacement) of the data, so no two trees see exactly the same examples.
Feature Randomness	At every node split, only a random subset of features is considered, preventing dominant features from monopolising every tree.
Parallel Prediction	Each tree independently outputs its own ROP estimate (e.g., 12.3 · 11.8 · 13.1 m/hr ...).
Averaging	The final prediction $ROP_{RF} = (1/N) \times \sum ROP_i$ averages all N tree outputs, reducing variance and smoothing outliers.

Architecture Diagram

Random Forest (RF) to Predict ROP



By combining hundreds of Decision Trees, Random Forest achieves higher accuracy and better generalisation than any single model — ideal for the complex, noisy nature of drilling data.

2 · K-Means Clustering

K-Means Clustering is an unsupervised machine learning algorithm that partitions drilling observations into K distinct groups (clusters) based on the similarity of their feature values. Observations within the same cluster share similar drilling conditions — such as comparable WOB, RPM, and flow rates — even if they come from different depths or time periods.

Why K-Means improves ROP prediction

Drilling conditions are not uniform: a soft sandstone formation drilled at 2 000 m behaves very differently from hard limestone at 4 500 m. A single global model struggles to capture all regimes simultaneously. K-Means helps by:

- **Segmenting the data** into homogeneous drilling regimes before training.
- **Adding a cluster label** as an extra feature, giving the Random Forest an explicit signal about which operational regime each observation belongs to.
- **Reducing within-group variance**, so the model learns sharper, more precise relationships inside each cluster.
- **Improving generalisation** — the model no longer needs one set of weights to cover wildly different conditions.

How K-Means works

Initialise	Choose K centroids randomly in feature space.
Assign	Assign every drilling observation to the nearest centroid (using Euclidean distance across normalised features).
Update	Recompute each centroid as the mean of all observations assigned to it.
Iterate	Repeat Assign → Update until centroids stop moving (convergence).
Label	Tag every observation with its cluster ID (0, 1, 2, ..., K-1) and pass this label as an input feature to the Random Forest.

Cluster Segmentation Example

Cluster 0	Cluster 1	Cluster 2	Cluster K
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Soft formation Low WOB High RPM	Medium formation Medium WOB Medium RPM	Hard formation High WOB Low RPM	... other drilling regimes
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K-Means transforms an undifferentiated mass of drilling records into structured groups, giving the Random Forest the contextual awareness it needs to predict ROP accurately across all formation types and operational conditions.

3 · Streamlit User Interface

Streamlit is a Python framework that converts data science scripts into interactive web applications with minimal code. In this project, Streamlit serves as the bridge between the trained AI model and the end user — a drilling engineer or operations manager who needs actionable insights without writing a single line of Python.

What the interface provides

Parameter Input	Sliders and numeric fields for WOB, RPM, Torque, Flow Rate, Depth (TVD & TMD), Lithology, and other drilling parameters.
Live Prediction	The moment the user adjusts any parameter, the trained Random Forest model computes and displays the predicted ROP in real time.
Cluster Display	Shows which K-Means cluster the current parameter combination falls into, helping the driller understand the operational regime.
Feature Importance	A ranked bar chart of the most influential features, so the driller knows which parameters to focus on for maximum ROP improvement.
Performance Metrics	Displays model accuracy metrics (R^2 , RMSE, MAE, MAPE) so users can judge how reliable the current prediction is.
Visualisations	Interactive charts showing predicted vs. actual ROP trends, cluster distributions, and historical drilling data.

Typical Workflow

- Step 1:** Driller opens the Streamlit dashboard in a web browser.
- Step 2:** Enters current drilling parameters (WOB, RPM, flow rate, depth, etc.).
- Step 3:** The app pre-processes inputs, applies the same feature engineering pipeline used during training.
- Step 4:** K-Means assigns the observation to the closest cluster.
- Step 5:** Random Forest generates the ROP prediction and displays it with confidence context.
- Step 6:** Driller adjusts parameters to explore how changes affect predicted ROP.
- Step 7:** Optimal parameters are identified and applied to the real drilling operation.

Streamlit turns a powerful machine-learning pipeline into a practical, real-time decision-support tool that any drilling professional can use without technical expertise.

4 · Feature Engineering

Raw drilling sensor data alone is insufficient for accurate ROP prediction. Feature engineering creates new variables that encode the physical interactions and relationships known from drilling mechanics. The table below lists every engineered feature used in this project.

Feature	Role
wob_rpm_ratio	Balance between WOB and RPM
wob_rpm_prod	Combined drilling energy
flow_rpm_ratio	Fluid circulation relative to RPM
flow_wob_prod	Interaction between cleaning and pressure
depth_tvd_wob	WOB effect at different depths
depth_tvd_rpm	RPM effect at different depths
depth_tmd_wob	Measured depth interaction with WOB
depth_tmd_rpm	Measured depth interaction with RPM
energy_proxy	Approximate drilling energy
wob_flow_ratio	Pressure relative to cleaning capability
rpm_flow_ratio	Rotation relative to fluid support
log_wob	Reduces extreme WOB variations
log_rpm	Reduces extreme RPM variations
log_flow_in	Reduces extreme flow variations

Outlier Handling

ROP values were clipped between the **1st and 99th percentiles** to remove extreme outliers caused by sensor malfunctions or exceptional geological events. This clipping improves model stability without discarding valid variation in the data.

The engineered features helped the model understand energy relationships, pressure interactions, depth effects, and fluid dynamics — the four pillars of drilling mechanics.

5 · Model Performance & Final Interpretation

Detailed Performance Metrics

Metric	Training	Testing
RMSE	6.86	8.46
R ² Score	86.06%	79.67%
MAE	4.05	5.25
MAPE	0.24	0.33

R² Score — 86% Training / 79.67% Testing

R² measures how much of the ROP variability the model explains. A training score of 86% shows the model strongly captured drilling patterns. A testing score of nearly 80% on completely unseen data confirms the model generalises well — an excellent result for noisy industrial data.

MAE — 4.05 Training / 5.25 Testing

Mean Absolute Error represents the average absolute difference between predicted and actual ROP. An MAE of ~5 ROP units on test data is acceptable given the inherent variability in geological formations, sensor readings, and operational conditions.

RMSE — 6.86 Training / 8.46 Testing

Root Mean Squared Error penalises large individual errors more heavily than MAE. The moderate RMSE values indicate that while some difficult drilling situations produce larger errors, the model remains stable overall. The RMSE-MAE gap is reasonable, confirming no systematic extreme mispredictions.

MAPE — 0.24 Training / 0.33 Testing

Mean Absolute Percentage Error of 33% on test data reflects the genuine difficulty of predicting a highly dynamic variable like ROP. As formation types, bit wear, and hydraulic conditions shift continuously, a sub-35% MAPE is considered acceptable in real-world drilling AI systems.

Training vs. Testing Gap

Training performance is slightly better than testing — this is expected and healthy. The gap is not excessively large, confirming the model is not memorising the training data (overfitting) but learning genuine

drilling relationships that transfer to new data.

Final Interpretation

The combination of Random Forest, K-Means Clustering, and rich feature engineering produced a system that is:

- **Accurate** – R^2 of ~80% on unseen data is strong for a noisy, high-dimensional problem.
- **Stable** – modest RMSE/MAE gap shows no extreme systematic errors.
- **Generalisable** – small train-test performance gap confirms real-world applicability.
- **Interpretable** – feature importance and cluster labels make predictions explainable.
- **Usable** – the Streamlit dashboard puts AI power directly in the driller's hands.

The model is capable of providing reliable ROP predictions and can serve as a practical AI-assisted drilling optimisation tool. The system successfully learns complex drilling relationships while maintaining good prediction accuracy on unseen operational data.